Arimoclomol for BUB1B-MVA

Can functional BubR1 be restored?

Conditional laboratory candidate

N1002K function and drug rescue remain unmeasured

The candidate must beat credible alternatives

Different evidence strengths, one unresolved functional question

Candidate	Strongest relevant support	Unresolved link
Arimoclomol	Human exposure record Conditional Hsp70 response	Functional mutant BubR1 rescue
NMN	Direct BubR1 increase MEFs and aged WT mouse testes	Mutant function and translation
4-PBA	Transient Hsp70 increase CF epithelial cells	BubR1 response and function

Same functional and fate endpoints. Candidate-specific exposure limits.

PMIDs 31900865, 24825348, 22069317 / MIPLYFFA and BUPHENYL labels

The decisive experiment

Full-length protein, matched abundance, measured function

1 **Establish the variant defect**

Full-length WT and N1002K at qualified matched abundance

2 **Compare function, exposure and cell fate**

Segregation errors with division, death and arrest denominators

3 **Apply a decision that can stop the lead**

Prespecified benefit bounds and independent confirmation

Open: laboratory acceptance, imaging qualification and complete costs

The software distinguishes three misleading signals

Recorded executions of the delivered Python decision rules

Protein rises 40%

Functional-benefit interval
−2 to +4 percentage points

Illustrative worthwhile benefit: 5 pp

Actual recorded Python output

WORTHWHILE_BENEFIT_RULED_OUT

Reason codes returned by Python

- BENEFIT_UPPER_BELOW_DELTA

Replication candidate IDs: [ ]

Recorded input_valid: true · process exit: 0

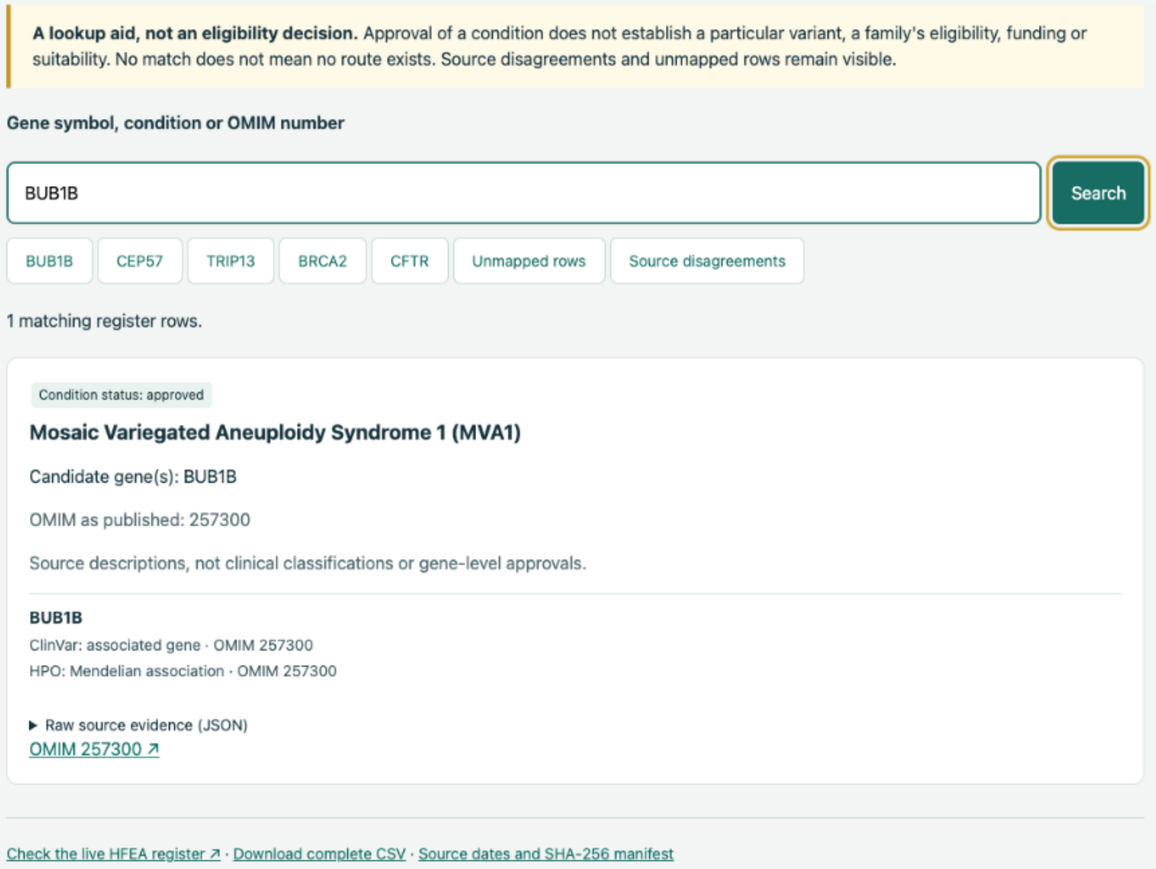
▶ **Criteria returned by Python**

▶ **Complete output · exact captured stdout**

SYNTHETIC INPUTS / supplied intervals / recorded Python executions

A reusable gene-to-condition navigator

Local queries with source evidence and unresolved cases visible



28 selected queries

47 source associations reviewed

Shared-source developer audit
Clinical validity untested

713 public benchmark signatures
Source-label recovery only

A credit-funded continuation with explicit outputs

$12,000 in Claude Science credits, conditional on award terms

Credit cap	Work	Concrete output
$3,600	Evidence checks	Corrected claim records
$3,000	Public validation	Protocol before inference
$2,400	Analysis tools	Reproducible releases
$600	Service qualification	Measured cost and suitability
$1,200	Reuse evaluation	Consenting reviewer feedback
$1,200	Reserve	Approved eligible follow-up

Laboratory funding and acceptance remain separate dependencies